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ORIGINAL ARTICLE

A bottom-up estimation of the heating and cooling demand in European industry

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Abstract Energy balances are usually aggregated at the level of subsector and energy carrier. While heating and cooling accounts for half the energy demand of the European Union's 28 member states plus Norway, Switzerland and Iceland (EU28 + 3), currently, there are no end-use balances that match Eurostat's energy balance for the industrial sector. Here, we present a methodology to disaggregate Eurostat's energy balance for the industrial sector. Doing so, we add the dimensions of temperature level and end-use. The results show that, although a similar distribution of energy use by temperature level can be observed, there are considerable differences among individual countries. These differences are mainly caused by the countries' heterogeneous economic structures, highlighting that approaches on a process level yield more differentiated results than those based on subsectors only. We calculate the final heating demand of the EU28 + 3 for industrial processes in 2012 to be 1035, 706 and 228 TWh at the respective temperature levels > 500 °C (e.g. iron and steel production), 100–500 °C (e.g. steam use in chemical industry) and < 100 °C (e.g. food industry); 346 TWh is needed for space heating. In addition, 86 TWh is calculated for the industrial process cooling demand for electricity in EU28 + 3. We estimate additional 12 TWh of electricity demand for industrial space cooling. The results presented here have contributed to policy discussions in the EU (European Commision 2016), and we expect the additional level of detail to be relevant when designing policies regarding fuel dependency, fuel switching and specific technologies (e.g. low-temperature heat applications).

Keywords Energy demand modelling · Temperature · End-use · Industry · Process · European Union

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Introduction

Energy balances are an important tool for researchers and policy makers alike, because they serve as a point of reference for analyses of the future energy system. However, the effort needed to generate primary data (for example recurring surveys of energy users) limits these balances to aggregated levels. Usually, they provide information on a sectoral and subsectoral level for individual or groups of energy carriers. The energy balance provided by Eurostat can be seen as a reference for Europe, since it applies a harmonised framework to the data collected by the national statistical institutes of the member states. While virtually all member states of the EU generate national energy balances, the availability of end-use balances is limited[1], especially in the industry sector. Currently, Eurostat does not provide end-use energy balances. The difficulties concerning the compatibility of national energy balances are described, for example, in the quality report of the European Union on energy statistics (European Commission 2014). EU regulation No. 1099/2008 and its amendments demand a 'greater focus (...) on final energy consumption' and more detailed and comparable energy statistics (European Parliament 2008). Thus, while not explicitly defining end-use dimensions, there is a clear need for data beyond the conventional dimensions of subsector and energy carrier. Additionally, more detailed energy balances could help to identify relevant applications for energy efficiency research by enabling more technology-focused approaches.

Strong heterogeneity among the countries of the EU28 + Norway, Switzerland and Iceland (EU28 + 3) can already be observed at subsector level (Fig. 1), as it is also supplied in the Eurostat energy balances (Eurostat 2016b)[2]. In terms of energy use, the most important sectors are iron and steel (22%, 533 TWh), chemical and petrochemical (19%, 448 TWh) and non-metallic minerals (15%, 353 TWh). The end-uses of this energy, and in particular the temperature levels of process heat, are not investigated in these energy balances. With the results of this paper, the heterogeneity highlighted here is explained on a more disaggregated level based on our bottom-up approach.

End-use balances break down the final energy demand into specified use categories as space and process heating/cooling, sanitary hot water and electrical appliances such as motors and lighting. For industry, only Germany, Switzerland, the UK and Austria provide more or less complete end-use balances with only Switzerland presenting data on all the investigated categories (space heating, hot water, appliances, process heating and cooling) (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016).

This lack of end-use balances in the EU is not only related to the general challenge of collecting detailed data on energy demand, but can be attributed in particular to industry-specific issues: evaluating process heat demand requires detailed knowledge about the characteristics of the various processes applied in European industry, and allocating temperature levels requires even more detailed information. And while individual processes, especially those with high overall or specific energy demand (energy-intensive industries), are well researched in terms of energy efficiency, the availability, comparability and reliability of the related data are an issue. Energy demand simulation models can be used to fill existing data gaps when complete data sets are required. The above-mentioned issues, however, mean that top-down approaches lack the desired level of detail to reflect industry's heterogeneity, while bottom-up approaches struggle with the availability of detailed technological and economic input data. Furthermore, some important aspects of industrial production like internal heat use and process integration cannot be represented at subsector level.

Previous work on this topic includes Pardo et al. (2013), who used similar approaches to the questions of how to assign a share of final energy for heating, or how to define temperature levels for industrial fuel use. They did so, however, with an emphasis on energy transformation from primary to useful energy for the industrial, residential and tertiary sector in the EU27. They present Sankey diagrams with the dimensions of subsector and temperature, stating distinct temperature profiles per subsector. Their results show that natural gas, petroleum products and coal products cover 83% of primary energy consumption. They supplement this result, which is in line with statistical data, by splitting the industrial useful energy demand into three temperature levels and the respective shares of useful energy in the EU27 (low 22%, medium 20% and high 58%). However, the absolute values provided are not directly comparable with Eurostat's energy balance, as they focus on useful energy.

Naegler et al. (2015) present two different approaches to disaggregating the final industrial energy demand of the EU28 for the year 2012 and conclude that an improved data basis is needed to achieve robust results. They calculate between 8150 and 8518 PJ industrial heat demand for the EU28, differentiated by five

[1] In the EU project 'Mapping and analyses of the current and future (2020–2030) heating/cooling fuel deployment (fossil/renewables)' (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016), a comprehensive analysis has been carried out of the availability of end-use balances in the EU28 + 3, including the residential and tertiary sector as well.

[2] The illustration shows model results for heating and cooling demand presented at the aggregation level also available in energy balances. The actual energy balances include additional electricity for non-heating/cooling uses.

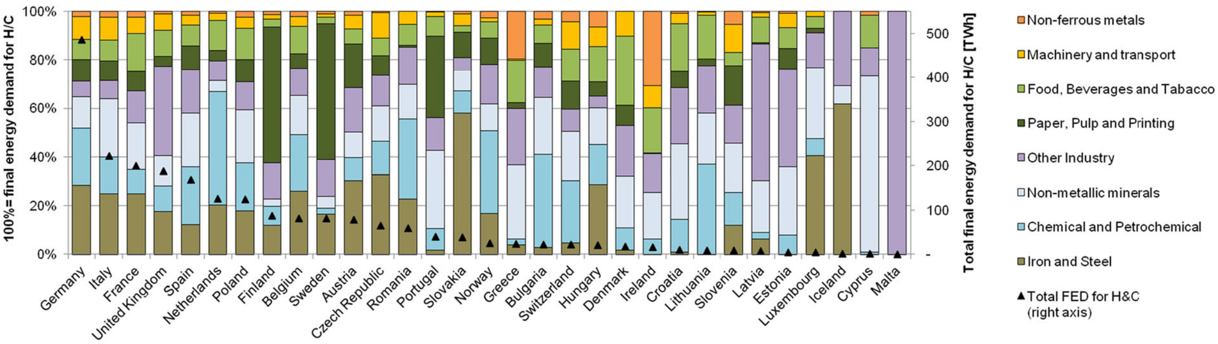


Fig. 1 Final energy demand for industrial heating and cooling by country and subsector in EU28 + 3 in 2012 (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016)

end-uses (space heating and hot water, four process heat temperature levels) and country. However, due to limited access to national bottom-up data, they applied Germany's end-use structure, as this was available in its national end-use balance. They assume that processes are the same across countries. They show that many approaches to the disaggregation of energy demand in the EU28 rely on similar data sets (e.g. subsector distribution of energy demand), thus producing similar results. This implies that different approaches are needed to substantiate or challenge the robustness of the available studies.

Both Pardo et al. (2013) and Naegler et al. (2015) assign a share of final energy to heating and cooling purposes and temperature levels, assuming similar shares across countries. However, both allocate this information at the level of subsectors and cannot account for substantial structural differences across countries within subsectors. For example, the iron and steel sectors in Italy and Germany feature significantly different shares of electric and oxygen steel.

We want to contribute to this line of research by introducing an approach that includes differences in national economic structures by incorporating process-specific information. This allows us to create technology-based temperature profiles for individual processes, which results in different temperature profiles for industrial subsectors in different countries due to their different economic structure. Thus, we reduce the dependency on national end-use balances that neglect country-specific differences when applied to the entire EU28 + 3.

In this paper, we present a methodology to disaggregate energy balances into end-use balances, with the focus on industrial heating and cooling demand by applying the bottom-up energy demand model FORECAST. We show how gaps in data availability are closed, and the bottom-up model results are connected to Eurostat's energy balance (Eurostat 2016b), enhancing it with the dimensions of end-use and temperature level.

In 'Methodology and Data' section, we present a description of the model and important data assumptions. In 'Results' section, we present selected dimensions of the model results, namely temperature distribution and energy carrier use, which are discussed in the last section. There, we also introduce the concept of 'bottom-up' coverage as a quality criterion. We conclude with our thoughts on how to proceed with the methodology and improve the results.

The results presented here were generated in a project for the European Commission and the full report and data sets (including non-heating/cooling use which is not presented here) are available online (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016).

Methodology and data

The model FORECAST-Industry (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016) works on the level of industrial processes, which are grouped into industrial subsectors and whose production figures vary among countries. The definitions of industrial subsectors and energy carriers are based on the energy balance of Eurostat. The model is divided into three parts, which are described following its workflow: industrial processes (including process heat and cold), space heating/cooling and bottom-up/top-down matching (Fig. 2). Processes and space heating/cooling are calculated using a bottom-up approach (top left/right), each based on an activity value and specific energy demand. The resulting bottom-up

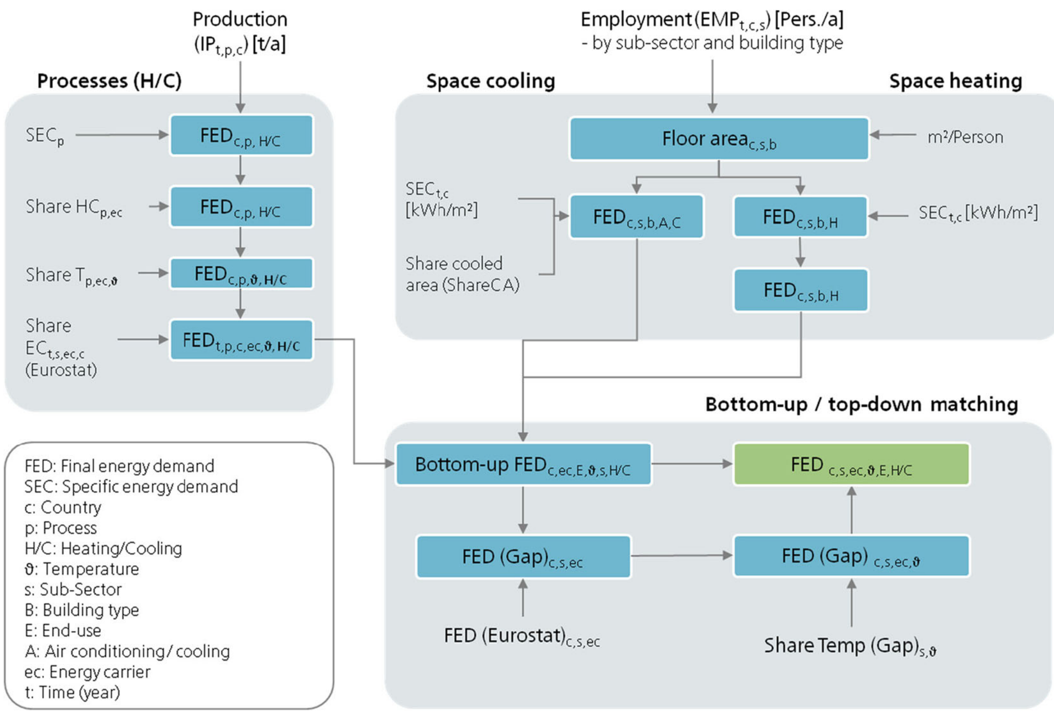


Fig. 2 Schematic representation of the end-use balance model calculation for the industry sector (FORECAST-Industry)

value is matched (bottom right) to top-down statistical values (e.g. Eurostat energy balance).

Processes

The heating and cooling energy demand of about 60 of the most energy-intensive industrial processes is calculated using a bottom-up approach as shown in Eq. (1) (see also Fig. 2). The considered dimensions are as follows:

- Countries (c) (European Union member states plus Switzerland, Iceland and Norway (EU28 + 3)
- Industrial subsectors (s) (iron and steel, chemical industry, non-metallic mineral products, paper and printing, non-ferrous metals, food, drink and tobacco), each with several industrial processes (p) assigned to them
- Energy carriers (ec) based on the energy balance of Eurostat 2016b). Some are grouped (ambient heat, solar energy, waste non-RES, district heating, biomass, coal, geothermal, waste RES, electricity, fuel oil, other fossil and natural gas). Note that the actual energy carrier share per country and subsector is taken from Eurostat (2016b) and not calculated bottom-up
- Temperature level (ϑ) (< 100, 100–200, 200–500, > 500 °C)[3]

$$FED_{p,c,\mathrm{ec},\vartheta} = IP_{p,c} \times SEC_p \times ShareEC_{s,\mathrm{ec},c} \times ShareH/C_{p,\mathrm{ec}} \times ShareT_{p,\mathrm{ec},\vartheta} \quad (1)$$

with FED as the final energy demand of a process, IP as its activity (production), SEC as the specific energy demand, ShareEC as the energy carrier share (taken at subsector level from the Eurostat energy balance), ShareH/C as the share of heating and cooling demand in the total energy demand and ShareT as the distribution of the processes' energy demand among the different temperature levels.

The industrial production per process and country IP in tonnes is based on several sources. A complete picture of the sources used is given in (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016). The actual values per process (sum of EU28+3) are given in Table 1.

Our definition of the temperature levels used is given in Table 2. Space heating is assumed to be supplied below 100 °C. An overview of the sources used for production in some important processes is presented in Table 3.

[3] The analysis presented here only comprises one base year, so there is no temporal dimension included. When used in scenario analysis, both the industrial production ($IP_{p,c}$) and the energy carrier shares ($ShareEC_{s,\mathrm{ec},c}$) may change each year.

Table 1 Investigated industrial processes, their specific energy consumption (SEC), process heating temperature distribution and the share of total electricity demand used for process heat

Subsector	Process	Energy demand		Share of electricity demand used for process heat (1)	Activity 2012 EU28 + NO,IS, H (Mt)	Temperature distribution (1)					Based on
		Fuels (GJ/t)	Electricity (GJ/t)			< 100 °C	< 100–200 °C	200–500 °C	500–1000 °C	> 1000 °C	
Iron and steel	Sinter	2.24	0.13	0.00	117.63	–	–	0.20	0.80	–	Arens et al. 2012, 2016
	Blast furnace	11.64	060	0.00	107.85	0.01	0.01	0.11	0.20	0.67	Arens et al. 2012, 2016
	Electric arc furnace	0.98	2.28	0.95	98.76	–	0.01	–	0.10	0.89	Arens et al. 2012, 2016
	Rolled steel	2.39	0.60	0.10	183.51	–	–	–	0.20	0.80	Arens et al. 2012, 2016
	Coke oven	3.20	0.12	0.00	46.33	–	–	–	0.20	0.80	Brauer 1996
	Smelting reduction	15.00	0.42	0.00	0.00	–	–	–	0.20	0.80	Hara et al. 1999, Primetals 2015
	Direct reduction	15.00	0.42	0.00	0.66	–	–	0.20	0.80	–	Midrex 2013, Cheeley 1999
Non-ferrous metals	Aluminium, primary	5.20	53.64	0.05	4.02	–	–	–	1.00	–	Fleiter et al. 2013, Winnacker-Küchler 2006
	Aluminium, secondary	9.00	1.67	0.30	3.27	0.28	–	0.30	0.42	–	Krone 2000
	Aluminium extruding	4.20	4.80	0.30	2.18	–	–	1.00	–	–	Krone 2000
	Aluminium foundries	7.20	5.60	0.30	2.43	–	–	–	1.00	–	Fleiter et al. 2013, Krone 2000
	Aluminium rolling	3.30	2.20	0.30	3.58	–	–	1.00	–	–	Krone 2000
	Copper, primary	8.00	2.79	0.20	2.08	–	–	–	–	1.00	Fleiter et al. 2013, Winnacker-Küchler 2006
	Copper, secondary	4.00	2.33	0.10	0.85	–	–	–	1.00	–	Winnacker-Küchler 2006
	Copper further treatment	2.00	3.78	0.15	5.20	–	–	1.00	–	–	Winnacker-Küchler 2006
	Zinc, primary	1.00	15.90	0.01	2.68	–	–	–	–	1.00	Winnacker-Küchler 2006
	Zinc, secondary	1.00	0.60	0.01	0.09	–	–	1.00	–	–	Winnacker-Küchler 2006
Pulp and paper	Paper	5.50	1.91	0.01	100.46	0.05	0.88	0.05	0.02	–	Fleiter et al. 2012, 2013, van Deventer 1997
	Chemical pulp	12.65	2.30	0.01	26.91	–	1.00	–	–	–	Fleiter et al. 2012, 2013, Bakhtiari et al. 2010
	Mechanical pulp	− 2.01	7.92	0.01	9.85	1.00	–	–	–	–	Fleiter et al. 2012, 2013, Laurijssen et al. 2012
	Recovered fibres	0.54	0.945	0.01	51.81	–	1.00	–	–	–	Fleiter et al. 2012, 2013, Laurijssen et al. 2012
Non-metallic minerals	Container glass	5.78	1.41	0.04	22.45	0.02	0.19	0.19	0.30	0.30	BREF Glass 2013, Büchel et al. 1999
	Flat glass	10.92	3.32	0.00	14.36	0.02	0.21	0.43	0.12	0.22	

Table 1 (continued)

Subsector	Process	Energy demand		Share of electricity demand used for process heat (1)	Activity 2012 EU28 + NO,IS, H (Mt)	Temperature distribution (1)					Based on
		Fuels (GJ/t)	Electricity (GJ/t)			< 100 °C	< 100–200 °C	200–500 °C	500–1000 °C	> 1000 °C	
											BREF Glass 2013, Büchel et al. 1999
	Fibre glass	4.92	1.81	0.20	2.70	0.02	0.19	0.19	0.30	0.30	BREF Glass 2013, Büchel et al. 1999
	Other glass	11.48	5.05	0.17	2.02	0.02	0.22	0.22	0.22	0.32	BREF Glass 2013, Büchel et al. 1999
	Houseware, sanitary ware	24.24	4.82	0.01	0.57	0.30	–	–	0.05	0.65	BREF Ceramic 2007, Büchel et al. 1999, Fleiter et al. 2013
	Technical, other ceramics	12.11	3.23	0.01	0.68	0.30	0.15	0.15	0.25	0.15	BREF Ceramic 2007, Büchel et al. 1999, Fleiter et al. 2013
	Tiles, plates, refractories	5.46	0.88	0.01	4.66	0.07	0.11	0.07	0.18	0.57	BREF Ceramic 2007, Büchel et al. 1999, Fleiter et al. 2013
	Clinker calcination-dry	3.50	0.14	0.00	172.06	–	–	0.10	0.60	0.30	Rahman et al. 2013, Patil and Khond 2014, Büchel et al. 1999
	Clinker calcination-semidry	4.00	0.16	0.00	10.34	–	–	0.10	0.60	0.30	Rahman et al. 2013, Patil and Khond 2014, Büchel et al. 1999
	Clinker calcination-wet	5.50	0.16	0.00	5.71	–	–	0.10	0.60	0.30	Rahman et al. 2013, Patil and Khond 2014, Büchel et al. 1999
	Gypsum	1.00	0.20	0.00	201.86	–	0.50	0.30	0.20	–	Bundesverband der Gipsindustrie 2013, Büchel et al. 1999
	Bricks	1.40	0.20	0.00	89.26	0.20	–	–	0.60	0.20	Fleiter et al. 2013, Dondi et al. 1997
	Lime burning	3.70	0.14	0.00	40.00	–	–	–	0.40	0.60	Gutierrez, Vandecasteele 2011, Büchel et al. 1999
Basic chemicals	Adipic acid	26.91	1.44	0.00	0.59	–	0.50	0.25	0.25	–	Fleiter et al. 2013, Weissermel, Arpe 1998
	Ammonia (synthesis gas)	11.27	0.48	0.00	18.00	–	–	–	0.66	0.33	Fleiter et al. 2013, Büchel et al. 1999
	Calcium carbide	6.12	8.32	0.95	0.34	–	–	–	–	1.00	

Table 1 (continued)

Subsector	Process	Energy demand		Share of electricity demand used for process heat (1)	Activity 2012 EU28 + NO,IS, H (Mt)	Temperature distribution (1)					Based on
		Fuels (GJ/t)	Electricity (GJ/t)			< 100 °C	< 100–200 °C	200–500 °C	500–1000 °C	> 1000 °C	
											Fleiter et al. 2013, Weissermel, Arpe 1998
	Carbon black	64.75	1.78	0.00	1.59	–	–	–	–	1.00	Fleiter et al. 2013, Büchel et al. 1999
	Chlorine, diaphragm	0.00	10.69	0.0	1.71	–	–	–	–	–	Fleiter et al. 2013, Büchel et al. 1999
	Chlorine, membrane	1.85	10.04	0.00	6.86	–	1.00	–	–	–	Fleiter et al. 2013, Büchel et al. 1999
	Chlorine, mercury	0.00	12.82	0.00	3.91	–	–	–	–	–	Fleiter et al. 2013, Büchel et al. 1999
	Ethylene	35.90	0.00	0.00	14.05	–	–	–	1.00	–	Fleiter et al. 2013, Weissermel, Arpe 1998
	Methanol (synthesis gas)	15.03	0.49	0.00	2.14	–	–	–	0.22	0.78	Fleiter et al. 2013, Weissermel, Arpe 1998
	Poly arbonate	12.86	2.66	0.00	0.85	–	1.00	–	–	–	Fleiter et al. 2013
	Polyethylene	0.64	2.04	0.00	8.94	–	1.00	–	–	–	Fleiter et al. 2013
	Polypropylene	0.79	1.15	0.00	8.64	–	1.00	–	–	–	Fleiter et al. 2013
	Polysulfones	24.49	3.06	0.00	0.36	–	1.00	–	–	–	Fleiter et al. 2013, Büchel et al. 1999
	Soda ash	11.33	0.33	0.00	9.82	0.30	0.40	–	–	0.30	Fleiter et al. 2013, Büchel et al. 1999
	TDI	26.69	2.76	0.05	0.60	–	1.00	–	–	–	Weissermel, Arpe 1998
	Titanium dioxide	34.23	3.34	0.00	0.45	–	0.30	0.23	0.35	0.12	Fleiter et al. 2013, Büchel et al. 1999
Food, beverages and tobacco	Sugar	4.50	0.71	0.00	17.69	0.10	0.60	–	0.30	–	Fleiter et al. 2013, Lauterbach et al. 2012
	Dairy	1.57	0.53	0.05	69.64	0.90	0.10	–	–	–	Fleiter et al. 2013, Lauterbach et al. 2012
	Brewing	0.97	0.39	0.05	39.36	0.55	0.45	–	–	–	Fleiter et al. 2013, Lauterbach et al. 2012
	Meat processing	2.04	1.55	0.05	58.48	0.40	0.60	–	–	–	Fleiter et al. 2013, Lauterbach et al. 2012
	Bread and bakery	2.40	1.45	0.45	24.65	0.20	0.33	0.47	–	–	Fleiter et al. 2013, Lauterbach et al. 2012

The specific energy consumption (SEC) is based on a variety of sources and process descriptions which are, due to the importance of this value, presented explicitly for each process in Table 1. The values give the total energy demand per tonne of product. They are not country-specific, as the analysed sources do not allow this differentiation. Instead, we have to assume that this basic process property does not vary significantly among the investigated countries. We call this assumption *process equals process*. Note that this is merely a simplification due to data availability for the purpose of the model. This constraint applies to all the technology data given in Table 1.

The energy carrier share in the respective subsectors ShareEC is not based on a bottom-up calculation but taken from the energy balances (Eurostat 2016b) as an average per subsector (each process uses the energy carrier distribution of its subsector). Thus, although it is country-specific, the energy carrier distribution cannot be investigated at process level.

The share of heating and cooling in total energy demand ShareH/C is based on the same sources as the specific energy consumption. It is mainly relevant for the share of electricity used in cooling (e.g. 96% of the electricity used in the oxygen production process is used for cooling), as fuels are considered to be used for heating purposes only (this may change if sorption cooling becomes more important).

The temperature profile ShareT of the processes is likewise based on the process descriptions given in Table 1 and shows which part of the total heating demand in a process is needed at which temperature level (e.g. 67% of the energy demand of blast furnaces is required above 1000 °C). To overcome the weakness of subsector-level temperature shares, we allocate the temperature shares and energy demand on the level of about 60 individual energy-intensive processes. This allows us to approach the processes using a technology-based methodology, which derives the required data from process descriptions based on the products produced and the technologies applied. The constraints of the *process equals process* concept apply.

Based on extensive literature analysis, we compile temperature profile and specific energy demand values (Table 1) of industrial processes. These process-specific data (SEC, temperature profile) aim to take the technical properties of the processes into account. These include economic considerations (i.e. which technology is applied the most) and auxiliary technologies (e.g. waste heat recovery, flue gas treatment). Analogies between individual process steps (e.g. common temperature profile elements when controlled cooling takes place (glass, ceramics) or when the product group inherently disallows very high temperatures (e.g. food products)) are used where possible and needed. The most important indicator is the highest temperature of the process, i.e. which temperature has to be achieved to create the final product. Medium-temperature ranges often occur in rather opaque process descriptions (e.g. glass slowly cooled to room temperature[4]). The majority of process descriptions were found in subsector-specific standard literature (Blüchel et al. 1999; Weissermel, Arpe 1998; Winnacker-Küchler 2006), others in previous attempts to assemble consistent process profiles (Fleiter et al. 2013), scientific articles on established and new process technologies (e.g. Arens et al. 2012, 2016; Hara et al. 1999; Cheeley 1999) and information material from companies where other sources were scarce (e.g. on direct and smelting reduction (Midrex 2013; Primetals 2015)).

Sometimes, the most relevant heat demand occurs not in the main production process but in necessary or beneficial side processes (e.g. drying residues in sugar production, preparing lye for chlorine production in the membrane process, producing synthesis gas in several chemical processes). In these cases, the process definition needs to be extended (e.g. as noted in Table 1, ammonia and methanol production includes synthesis gas as a process step). In general, processes and products with high macroeconomic relevance are documented the most thoroughly (e.g. steel, cement, bulk chemicals), while smaller product groups (e.g. special ceramics) are harder to capture due to their heterogeneity. Processes with relatively low-energy intensity (e.g. food) are also less well documented, probably because the manufacturer's focus here is not on energy. Sources for these processes are instead often characterised by a technology-centred approach (e.g. supply of low-temperature process heat via solar thermal systems (Lauterbach et al. 2012)).

As some of the sources used are quite old, it may be argued that the values are outdated. And while we constantly update the process database where possible, we address this argument with several measures and assumptions because new data cannot be generated at will:

[4] It may be unclear, for example, whether this cooling is carried out actively (with additional energy demand or waste heat use) or passively.

Table 2 Definition of temperature levels for industrial process cooling and process heating

End-use	Temperature level (°C)	Comment
Process cooling	< − 30	Mostly air separation in chemical industry
	− 30–0	Mostly refrigeration in food industry
	0–15	Mostly cooling in food industry
Process heating	< 100	Low-temperature heat (hot water) used in food industry and others
	100–200	Steam, mostly used in paper, food and chemical industry
	200–500	Steam, mostly used in chemical industry
	> 500	Industrial furnace in steel, cement, glass and other industries

– The most energy-intensive processes usually experience evolutionary efficiency improvements (in contrast to revolutionary). One part of FORECAST simulates these incremental energy savings. It is not presented here but the included efficiency measures are discussed in Fleiter et al. (2013).
– We assume that the temperature distribution of processes does not change significantly over time.
– The values presented here do not reflect the best available technology (BAT) but try to capture the actual average stock, which is slower to change.
– Finally, as the actual SEC does vary from the values presented in Table 1, we emphasise the concept of *bottom-up coverage* (see below) as a quality criterion. If energy efficiency increases, this value will change and indicate sectors that need updated data.

Thus, while we are constantly searching for up-to-date data, the model does show some resilience regarding outdated specific energy consumption data.

Our definition of the temperature levels used is given in Table 2. Space heating is assumed to be supplied below 100 °C.

In order to include temperature levels, we assume that all fuels are used to produce heat, stated in ShareH/C[5] in (1), while the major part of electricity (estimated to be around 90%) is used in non-heat applications. Exceptions to this are electric arc furnaces in steel and calcium carbide production and some electric melting furnaces. We assume that other applications for fuel use (like direct mechanical energy) are negligible. On-site generation of electricity is considered part of the transformation sector in line with the Eurostat energy balance's definition. However, the model does include the heat produced in small combined heat and power installations.

Non-heat energy use (e.g. in cross-cutting technologies such as motor appliances or lighting) is tracked to complete the energy balance, but not presented here. This non-heat energy use consists solely of electricity (see definition of ShareH/C).

Process cooling

Process cooling is used in industry for different purposes. In Europe today, the majority (over 90%) of refrigeration systems in industry are based on compression technology powered by electricity. Only stationary cooling units are analysed; mobile cooling applications are not included. It is assumed that production processes in the individual European countries do not differ significantly in their characteristic properties for the production of process cold; *process equals process* (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016).

In contrast to the process heating side, there is hardly any statistical information available on process cooling generation in Europe. Scientific studies are also rarely available (European Commission 2007; European Commission 2011; SVK 2012; VDMA 2011; UBA 2015). There are large data gaps for process cooling applications (number of units, capacity of refrigerating plants, energy efficiency, information about maintenance, service etc.) in the various European countries. Technology stock data is rather limited and is partially found in EcoDesign preparatory studies (European Commission 2007; European Commission 2011). The

[5] The share of energy demand used for heating and cooling (ShareH/C) translates the total energy demand given by the SEC into process heat demand.

stock of refrigeration systems remains not at a constant level over the years. Its development depends on the assumed cooling demand and the lifetime of existing and new systems. After reaching the maximum lifespan, existing refrigeration systems leave the stock. New refrigeration systems are added as required. In industry, the assumed lifetime for calculations is 20 years; however, the lifetime of these systems depends also on the size of the installed devices (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016). In the services sector, the average lifetime was estimated to be 15 years. Other estimates of typical lifetime spans of domestic refrigeration systems are 12–20 years, of transport refrigeration systems 6–10 years and of chillers 15–30 years. The real lifetime of refrigeration systems depends on various influencing factors, not only the size of the system. Therefore, data on lifetime of refrigeration systems should only be taken as estimates.

Process cooling data are not recorded anywhere, and the area of process cooling is not represented in Eurostat statistics. This is why the relevant associations in Europe and at member state level as well as cooling experts in some countries (Germany, Switzerland, the UK, Austria and Belgium) were contacted directly with the objective of collecting public and non-publicly available data. A formal data request including a template of the desired information was sent to over 25 different institutions in Europe (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016). The response rate was reasonable (over 50%), but the requested data was non-existent in almost all cases.

Furthermore, there are very individual cooling process solutions for different types of clients and aggregated information has not been collected, except in some member states where studies have been conducted at sector level on energy use and the demand for process and space cooling (e. g. Switzerland). We undertook in-depth desk research and analysis of specific cooling studies in Germany, Switzerland and the UK. On several occasions, the authors of these studies (cooling experts) were contacted directly as were the producers of particular chemicals (gases) with an especially high demand for cooling.

Based on the high complexity and the lack of resilient data concerning process cooling in European industry, it was necessary to make expert-based technical assumptions (stock of refrigeration plants, average capacity size for cooling systems (per branch), typical full load hours per branch, specific energy demand for process cooling per energy-intensive industrial process, allocated energy for process cooling, analysis per branch etc.). For example, different full load hours of cooling devices were assumed depending on the average temperature in the various countries. In order to validate the assumptions for the calculation of the technology stock (e.g. full load hours, average installed capacity), interviews were conducted with organisations and experts. These assumptions were included when calculating the energy demand for process cooling in European industry. Therefore, the results contain uncertainties that are estimated to amount to ± 25%. The indication of such a range was discussed with cooling experts. All experts considered such an indication to be useful in order to accommodate the uncertainties in the cooling market and the cooling applications.

Space heating/cooling

Similar to the calculation of process energy demand, but on a subsectoral level, the final energy demand for space heating and cooling in industry (FED) results from the activity, employees (EMP) (Eurostat 2016a) and specific energy demand per square metre of floor area (SEC). The latter is adjusted to countries using the European heating/cooling index EHI/ECI described by Werner (2006, 2015), (EH(C)I). He states that the specific energy demand is proportional to the square root of the heating/cooling degree days, rather than directly proportional. Tables 3, 4 and 5 show representative values of floor area per employee (EMPArea) and SEC. The final energy demand is calculated individually for heating

Table 3 Sources of industrial production figures

Subsector/process	Data source
Iron and steel	World Steel Association 2016
Cement	Cembureau 2013
Glass	Glassglobal 2017
Pulp and paper	German Pulp and Paper Association (VDP) 2016, UNdata 2017 (FAO)
Aluminium and copper	US Geological Survey 2017
Chemicals: ammonia	UNFCCC 2017
Chemicals: ethylene	UNFCCC 2017
Chemicals: oxygen	Eurostat
Chemicals: methanol	UNFCCC 2017
Chemicals: chlorine	Eurochlor 2017
Food, drink and tobacco	UNdata 2017 (FAO)

Table 4 Area per employee in square metre by industrial subsector and building type (assumptions based on Biere (2014))

m^2 employee	Building type	
Subsector	Production	Office
Iron and steel	82	67
Non-ferrous metals	82	68
Paper and printing	82	123
Non-metallic mineral products	82	124
Chemical industry	82	112
Food, drink and tobacco	82	174
Engineering and other metal	67	57
Other non-classified	82	65

and cooling but shares the data input of heated/cooled area (EMP, EMPArea)[6].

$$\mathrm{FED}_{c,s,b} = \mathrm{EMP}_{c,s} \times EMPArea_{c,s,b} \times \mathrm{SEC}_b \times \mathrm{EH(C)I}_c \times ShareCA \quad (2)$$

As there are no comprehensive statistics about floor area in industrial subsectors, we calculate this based on employment in each subsector (EMP) (Eurostat 2016a) and the specific floor area per employee (EMPArea) (Biere 2014).

The specific energy consumption for heating purposes (Table 5) differentiates two building types and seven age classes (Biere 2014).

The European heating (and respectively cooling) index EHI, ECI[7] (Werner 2015) introduces differences in SEC among countries.

For cooling, we include an additional estimation about the share of cooled floor area (ShareCA)[8].

Indices depict country (c), building type (b) and subsector (s). Since data on space heating and cooling in industry are generally scarce, the results are calibrated[9] using top-down values given by Eurostat (2007), while retaining subsectoral information about the distribution.

[6] The analysis presented here only comprises one base year, so there is no temporal dimension included. When used in scenario analysis, both the employment data ($\mathrm{EMP}_{c,s}$) and the specific energy demand of the buildings ($\mathrm{SEC}_{c,b}$) may change each year.
[7] See supplementary data for values.
[8] See supplementary data for values.
[9] See supplementary data for values and details on data origin.

Table 5 SEC for industrial space heating by construction year of building and building type (Biere 2014)

Construction year	Production buildings SEC in kWh/m^2	Office buildings SEC in kWh/m^2
1950–1959	243	270
1960–1969	243	240
1970–1979	243	180
1980–1989	213	140
1990–1999	151	120
2000–2009	90	100
2010–2019	29	55

Results

We present our disaggregated energy balance of heating and cooling demand, focusing on the dimensions added to the Eurostat energy balance: temperature level and application (process heat, space heat, cooling). The heterogeneity among subsectors already presented in Fig. 1 is accompanied by heterogeneity at process level, which we present for the iron and steel industry in Fig. 3. It shows the share of energy used by individual processes in the subsector. Note that this is a result of the bottom-up calculation and as such not calibrated. The earlier mentioned restrictions regarding bottom-up coverage apply to this (and each result presented at process level). The very heterogeneous distribution of process shares highlights the fact that the results benefit from process differentiation among countries and that the assumption of identically distributed subsectors may not be suitable for all subsectors and countries. This is amplified in subsectors with many processes and products, like non-ferrous metals (Fig. 4).

Temperature distribution

Of the 1035 TWh of process heat above 500 °C in the EU28 + 3, 96% are used in the three subsectors iron and steel, basic chemicals and non-metallic minerals, with iron and steel alone contributing 48% (493 TWh) (Fig. 5). Basic chemicals and non-metallic minerals follow with 25% (260 TWh) and 23% (240 TWh), respectively. This is mainly caused by the combination of high absolute energy demand and high temperature level, which is found in the following processes: blast furnaces, sinter, coke ovens and electric arc furnaces (iron and steel), clinker burning (non-metallic minerals) and ammonia and ethylene production (basic chemicals). The finding that these processes contribute the most to the

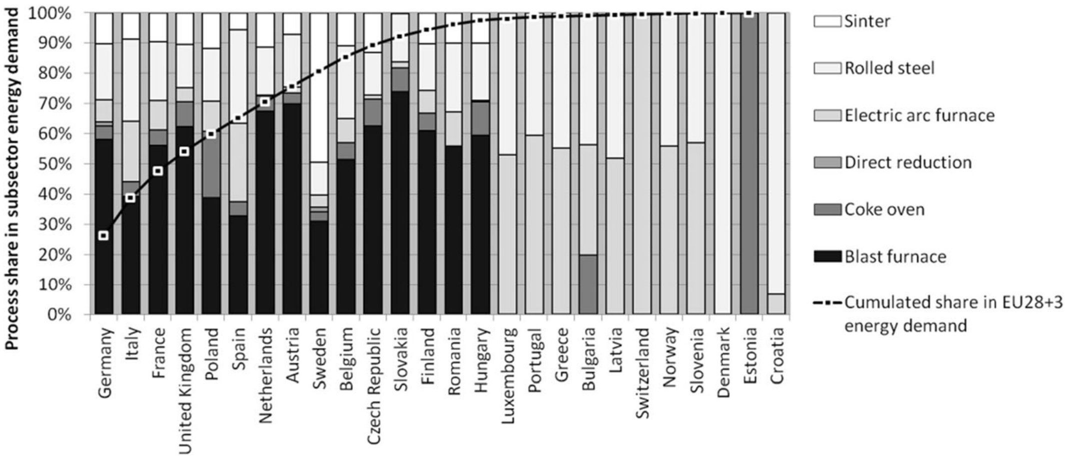


Fig. 3 Share of processes in the iron and steel subsector FED by country, cumulated share of country's FED in EU28 + 3 FED

overall high-temperature demand is consistent for the EU28 + 3 as a whole. Note that this still varies for individual countries; there are countries, for example, that do not engage in blast furnace operations (see also Fig. 3). Process heat demand between 100 and 500 °C, which we assume to be the typical temperature range for steam systems, is most prevalent in the paper and printing subsector (217 TWh) as well as in other industries (203 TWh) that were not subject to detailed modelling at process level and whose results are therefore less reliable. Notable other subsectors in this respect are non-metallic minerals (82 TWh), food, beverages and tobacco (78 TWh) and basic chemicals (51 TWh). Low-temperature (< 100 °C) process heating (228 TWh), characterised by the use of hot water, is mostly used in chemicals (34%, 78 TWh); food, beverages and tobacco (27%, 61 TWh); paper, pulp and printing (13%, 29 TWh) and other industries (11%, 26 TWh).

In absolute terms, process and space cooling are almost negligible compared to heat demand (about 100 TWh compared to about 2300 TWh). Approximately 75% of the total cooling system stock (about 73,000 cooling systems) in industry is located in seven countries: Germany, France, Italy, the Netherlands, Poland, the UK and Spain. Space cooling is mainly used in the Mediterranean countries of Italy, France, Spain, Portugal and Greece, and only to a very low extent in the northern European countries. However, there are significant differences concerning the process cooling demand among industrial sectors, as presented in Fig. 6: Very low temperatures (< − 30 °C) occur mainly in basic chemicals (20 TWh) in the context of air separation. A very small proportion of low-temperature cooling demand is found for example in research and development processes of different industry branches. Notable other cooling demand ranging from 0–15 °C (66 TWh) exists in food,

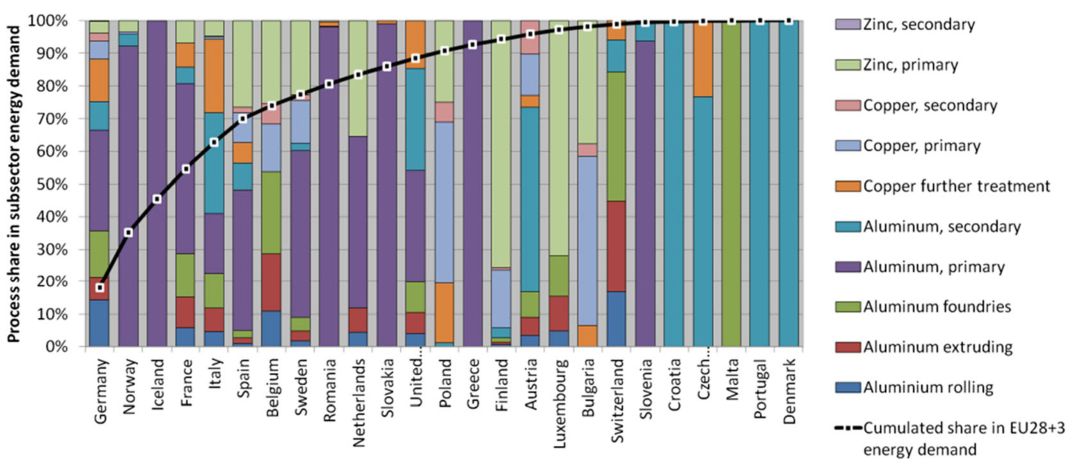


Fig. 4 Share of processes in the non-ferrous metals industry FED by country, cumulated share of country's FED in EU28 + 3 FED

beverages and tobacco (46 TWh), with the other subsectors combined adding up to only 19 TWh. Space cooling amounts to 12 TWh (see also Table 6).

By adding the process dimension, we can investigate the most important processes in terms of a high share of bottom-up-explained energy demand in total top-down energy demand. To define 'high share', we arbitrarily set a limit of at least 1% of the bottom-up explained heating demand (which is approx. 90% of total top-down energy demand, 2078 TWh)). This is true for 25 individual processes which together make up a total of 87% of the BU-explained energy demand. Among the other processes are those with low specific energy demand (e.g. primary and secondary zinc production), low activity (e.g. semidry clinker calcination) or those mainly using electricity in applications like mechanical energy (e.g. mechanical pulp) and electrolysis[10]. Figure 7 shows the most relevant processes in terms of absolute energy demand with their bottom-up (BU) process heat demand and temperature distribution. We can observe that processes with high-energy demand (absolute and specific) also tend towards high temperatures (> 500 °C)[11], with the notable exception of paper production, which shows rather average SEC (around 7 GJ/t) but high activity as a mass product. In the process, mostly steam and hot air of medium temperature are used to dry the paper rolls. Steel production, especially the blast furnace route (see e.g. Arens et al. 2016, Worrell et al. 2008 for technology descriptions), dominates high-temperature demand due to both its high specific energy demand (around 20 GJ/t crude steel) and high production quantities (170 Mt in the EU28 2014, (World Steel Association 2016)). Other important processes belong to the chemical, non-metallic minerals, non-ferrous metals and food industries.

If we turn away from processes and focus on end-uses, we can observe a relatively stable temperature distribution among the larger countries (Fig. 8 and Table 6). Robust findings include high-temperature shares (process heating above 500 °C) between 40 and 50%, which are found in 13 countries and account for approx. 80% of the total energy demand for heating and cooling (H/C). Medium-temperature (200–500 °C) energy demand between 5 and 10% can be observed for 21 countries and 86% of the total H/C demand. There are stronger variations in process heating shares between 100 and 200 °C and below 100 °C, which range from 4 to 50% and 4 to 29%, respectively. Notable exceptions are Iceland, Cyprus and Malta, which show remarkably high-relative cooling demand shares. This is probably caused by low data availability, sometimes limited to electricity or certain subsectors in terms of top-down energy demand[12]. This also applies to activity data, limiting the reliability of our results for these countries in general. Another conspicuous result is the high share of process heating between 100 and 200 °C in Sweden and Finland. This can be explained by the high importance of the pulp and paper industry in these countries, accounting for 33% (58 TWh) and 27% (50 TWh) of their total heating demand, respectively. Slovakia shows a high share of high-temperature energy demand, which is caused by the iron and steel industry with 21.9 TWh above 500 °C (54% of the country's total reported industrial energy demand). The same is true for Luxembourg (3.1 TWh above 500 °C, 53% of the reported total). Compared to the EU28 + 3, Germany seems to be a good proxy regarding overall end-use; but country-specific structures can still have a strong impact.

Energy carriers

According to our model results[13], the most frequently used energy carriers for industrial heating and cooling (Fig. 9) are natural gas (935 TWh, 39% of total), coal (415 TWh, 17%), other fossils (243 TWh, 10%), biomass (216 TWh, 9%), fuel oil (208 TWh, 9%), district heat (184 TWh, 8%) and electricity (173 TWh, 7%). Germany, Italy, France, the UK and Spain account for 57% (1374 TWh) of the

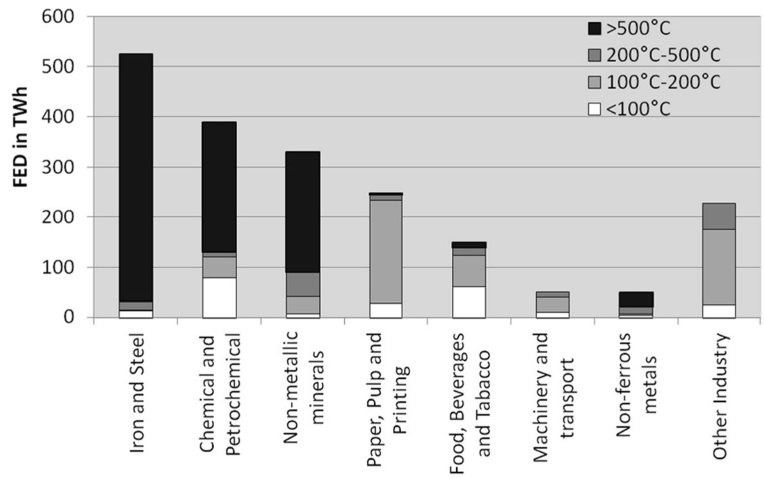
[10] For electrolysis, we included shares of electricity use as process heat, depending on the specific process description. Since, for example, electrolysis in the primary production of aluminium in the Hall–Héroult process requires a temperature of around 950 °C, we assign approx. 2.5-GJ/t electricity use as process heat. While the theoretical minimum would be approx. 0.8 GJ/t, special conditions of the process induce heat losses (US Department of Energy 2008; Nowicki and Gosselin 2012). The distinction between process heat in our definition and process-specific electricity use is therefore not trivial.

[11] Of the 25 processes presented, the ten biggest include eight processes that mainly or exclusively use high-temperature heat above 500 °C.

[12] For example, for 2015, Eurostat's energy balances (Eurostat 2016b) show 46 ktoe (0.53 TWh) industrial energy demand for Malta, with 36 ktoe (0.42 TWh) of electricity. This proportion is very different from the ones we observe in other economies. The model is not well suited to deal with this, whether the cause is a real difference in economies or a statistical issue.

[13] The general distribution of energy carriers can already be derived from Eurostat (2016b). The main difference in these results is that they refer to heating and cooling instead of total energy demand. This does, however, mainly affect electricity, as fuels are assumed to be used for heating uses only.

Fig. 5 Industrial process heat demand by subsector and temperature level, EU28 + 3



total heating and cooling demand of the EU28 + 3 (2413 TWh), while the 14 smallest countries in terms of energy use together account for 10% (253 TWh).

FORECAST uses an average energy carrier share per subsector and country (i.e. not per process), based on the respective shares in the subsector's energy balance (Eurostat 2016b). The means that the subsector's energy carrier distributions are automatically meet the energy balance. At the same time, the energy carrier use cannot be analysed on process level. Thus, for Fig. 10, it has to be noted that the temperature distribution refers to subsector-use of energy carriers. We can still observe that the distribution of energy carriers to temperature levels generally confirms expectations: Coal and electricity are used for high-temperature energy demand (steel industry, electrolysis), natural gas is used in all temperature levels and biomass is used in low-temperature subsectors (e.g. paper production). The share of high-temperature use of waste is most likely connected to its use in clinker production.

Discussion

Results compared to Eurostat energy balance: bottom-up coverage

The bottom-up results for space and process heating/cooling (plus non-heat use) do not match the energy balance (Eurostat 2016b) in most cases, because of general data imperfections and because not all industrial processes are covered. We call the degree to which the bottom-up (BU) values match the top-down (TD) energy balance '*bottom-up coverage*'

Fig. 6 Industrial process cooling demand by subsector and temperature level, EU28 + 3

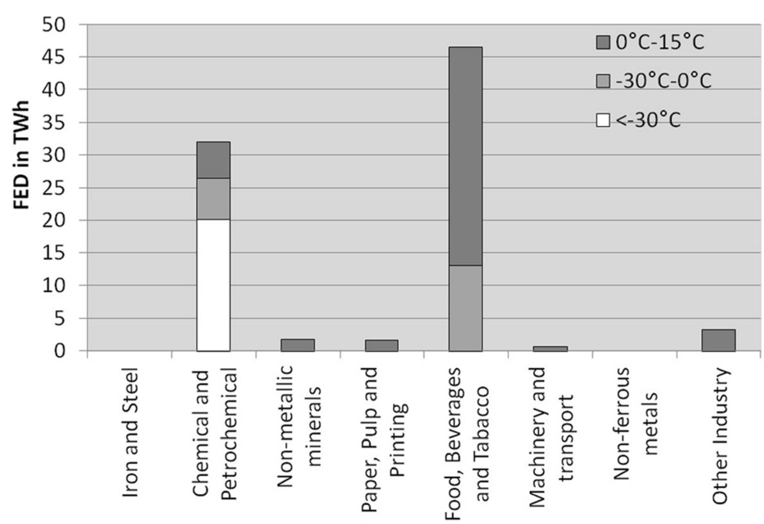


Table 6 Industrial process and space heating/cooling demand by country and temperature level EU28 + 3 (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016)

TWh	Space heating	Process heat < 100 °C	Process heat 100–200 °C	Process heat 200–500 °C	Process heat > 500 °C	Heating	Process cooling < − 30 °C	Process cooling − 30–0 °C	Process cooling 0–15 °C	Space cooling	Cooling	Heating and cooling
Austria	13.4	3.3	22.0	5.3	35.5	79.4	0.7	0.4	0.7	0.1	1.9	81.3
Belgium	13.9	7.1	14.8	6.7	42.4	84.8	1.6	0.9	1.9	0.1	4.5	89.3
Bulgaria	5.4	5.1	3.5	1.5	6.9	22.5	0.3	0.2	0.4	0.2	1.0	23.5
Switzerland	2.0	2.8	6.1	2.2	9.6	22.7	0.1	0.3	0.6	0.0	1.0	23.7
Cyprus	0.1	0.1	0.2	0.2	0.9	1.5	0.0	0.0	0.1	0.0	0.1	1.7
Czech Republic	9.4	7.7	13.1	5.5	30.3	66.0	1.0	0.4	0.8	0.1	2.3	68.3
Germany	58.3	64.3	92.1	34.6	251.1	500.3	5.3	3.9	9.2	0.1	18.5	518.8
Denmark	3.5	3.1	5.6	1.8	4.3	18.4	0.0	0.3	0.8	0.0	1.2	19.6
Estonia	1.6	0.4	1.3	0.4	0.9	4.5	0.0	0.0	0.1	0.0	0.2	4.7
Greece	4.2	2.7	5.6	2.0	9.7	24.2	0.1	0.2	0.5	0.7	1.4	25.6
Spain	31.4	4.4	45.3	14.5	82.3	177.8	1.1	1.5	3.8	3.5	9.9	187.7
Finland	6.2	16.1	45.7	5.3	15.6	89.0	0.8	0.4	1.1	0.0	2.2	91.2
France	46.8	8.6	38.5	15.8	97.7	207.3	0.7	2.7	6.3	1.7	11.3	218.7
Croatia	1.3	1.4	2.9	1.1	3.6	10.3	0.0	0.1	0.2	0.1	0.4	10.7
Hungary	4.7	3.8	2.3	1.1	9.3	21.2	0.1	0.2	0.4	0.1	0.8	22.0
Ireland	3.9	1.4	4.5	1.7	5.3	16.8	0.0	0.3	0.6	0.0	1.0	17.8
Iceland	0.4	0.1	0.1	0.0	1.5	2.2	0.0	0.0	0.3	0.0	0.3	2.5
Italy	47.7	27.9	26.5	16.6	117.7	236.4	3.2	2.0	5.2	4.5	14.9	251.3
Lithuania	1.7	2.7	1.7	0.6	2.2	8.8	0.0	0.1	0.2	0.0	0.3	9.2
Luxembourg	0.6	0.1	0.5	0.7	3.9	5.7	0.0	0.0	0.1	0.0	0.1	5.8
Latvia	1.4	0.5	3.0	1.0	2.2	8.2	0.0	0.0	0.1	0.0	0.1	8.3
Malta	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1
Netherlands	14.9	17.4	21.1	7.6	66.5	127.6	2.3	1.2	2.4	0.1	5.8	133.4
Norway	5.6	0.5	5.0	1.1	14.8	27.0	0.2	0.5	0.9	0.0	1.5	28.6
Poland	10.0	15.4	29.5	10.7	62.0	127.6	1.4	1.1	2.4	0.1	4.9	132.5
Portugal	6.7	4.0	15.0	2.5	13.1	41.4	0.1	0.3	0.6	0.6	1.7	43.0
Romania	10.4	4.0	8.5	4.0	33.1	60.0	0.4	0.3	0.7	0.0	1.4	61.4
Sweden	7.6	4.9	49.6	7.0	15.4	84.4	0.1	0.4	1.0	0.0	1.5	85.9
Slovenia	1.7	0.6	2.2	0.5	3.6	8.6	0.0	0.0	0.1	0.0	0.2	8.8
Slovakia	6.1	1.7	4.0	0.9	26.7	39.4	0.4	0.1	0.2	0.1	0.9	40.4
UK	25.6	16.4	61.0	22.1	66.3	191.3	0.3	1.7	4.5	0.0	6.6	197.9
EU28 + 3	346.3	228.1	531.2	175.3	1034.6	2315.6	20.1	19.4	46.1	12.4	98.0	2413.6

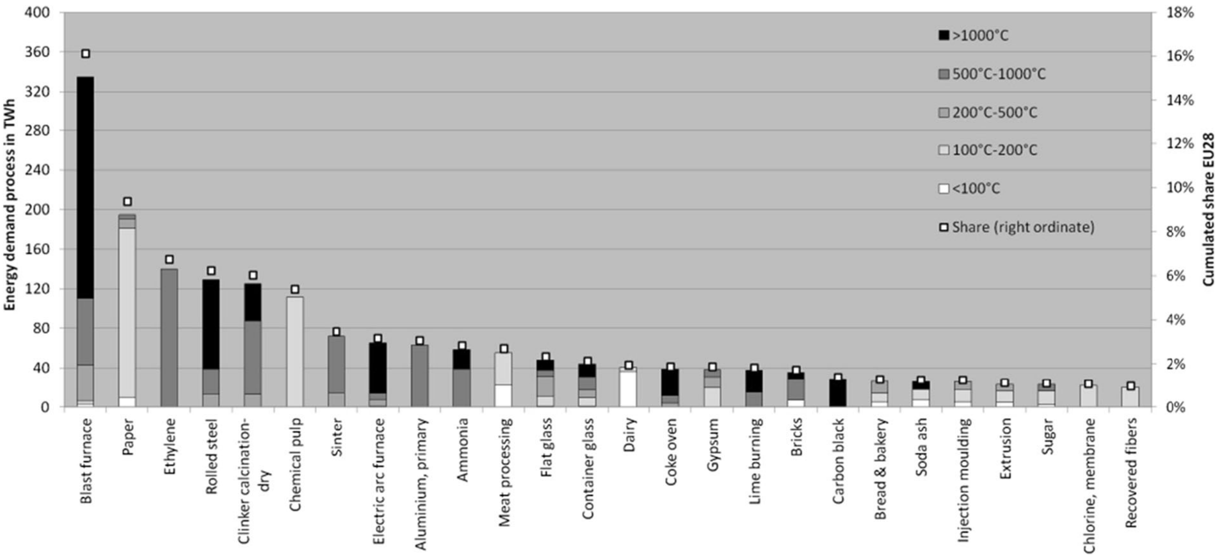


Fig. 7 Selected industrial processes, their BU process heat demand, temperature distribution and share in total industrial BU process heat (right ordinate) in the EU28 + 3

(BU-coverage). A value of '1' means that the BU-values are in line with the energy balance. Lower values indicate energy demand which is not accounted for in the BU-calculation. We assume that this gap occurs mainly due to energy use in processes not covered in the model. Subsidiary temperature distributions at subsector level (Fig. 11) are assigned to this energy use. Since these distributions lack the level of detail of the BU-calculation, achieving high BU-coverage is a quality criterion of the model.

Bottom-up coverage values greater than 1 can occur if the SEC or the production of processes is overestimated. In these cases, the respective overestimated processes might be overrepresented in their subsector.

The dimensions of the BU-coverage are subsectors (e.g. iron and steel, chemical industry) and countries (Figs. 12, 13, 14 and 15). Note this includes only heating- and cooling-related energy demand (i.e. excludes electricity use for lighting, mechanical energy etc.). The aggregated BU-coverage (across countries or subsectors) is calculated using final energy demand (FED)-weighted averages of the lowest level of aggregation (individual subsector and country)[14]. In general, the energy demand in the iron and steel subsector is somewhat overestimated (Figs. 12 and 15). This is most likely related to the use of derived gases (e.g. stack gas, converter gas) in power plants and to autoproducers of electricity, whose balance in Eurostat and national energy balances is not necessarily compatible with technology-focused bottom-up estimations. The chemical industries are underestimated because several processes here are too small to include in FORECAST. The BU-values of 'Other non-classified' and 'Engineering and other metal' do not include processes for the same reason, so their BU-values consist solely of space heating (i.e. no bottom-up processes are calculated for these subsectors). In Figs. 12, 13, 14 and 15, the right ordinates show the cumulated share of the displayed categories. Figure 13 reveals that the apparently good average coverage for the 'Engineering and other metal' subsector on the EU28 + 3-level in Fig. 12 is a rather random result. In fact, many countries, among them, the three biggest energy users in this subsector (Germany, Italy, France), which together account for 56% of the EU28 + 3's energy demand, show a BU-coverage of around 0.5, while some smaller countries' contributions in this subsector are overestimated. As our methodology relies on technology-based process descriptions, it yields little or no benefit in subsectors that are characterised by a huge variety of heterogeneous products like the engineering sector. In total, the energy demand in subsectors not readily accessible to our approach amounts to 700 TWh or 22% of the industrial energy demand in the EU28 in 2014 (Eurostat 2016b).

[14] This means that countries' and subsectors' BU-coverage is weighted by the final energy demand (FED) of the respective aggregated dimension: Countries with low FED contribute less to BU-coverage in Fig. 4; subsectors with low FED contribute less to BU-coverage in Fig. 6.

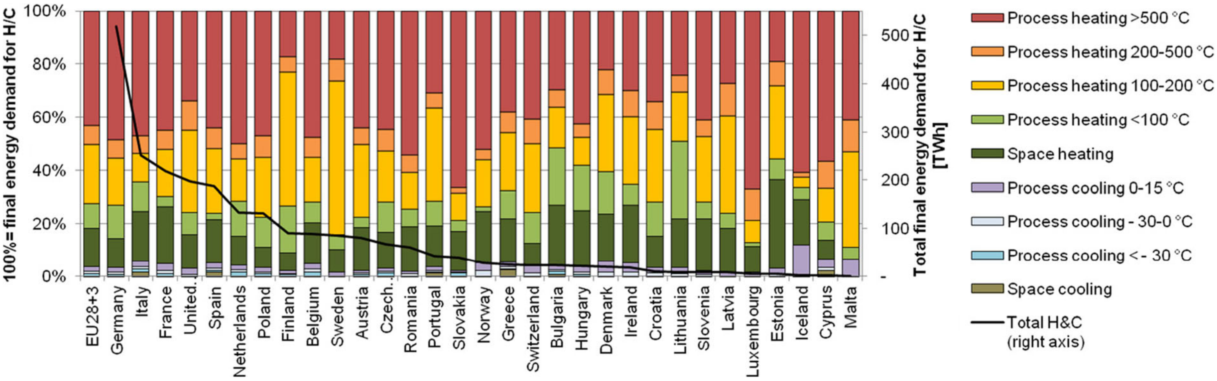


Fig. 8 Temperature share and final energy demand for industrial heating and cooling by country for the EU28 + 3 (Fraunhofer ISI, Fraunhofer ISE, TU Wien, TEP Energy, IREES, Observer 2016)

We observe that the seven biggest countries, accounting for approx. 65% of FED, have a BU-coverage of about 90% (Fig. 14), which we consider a good value (100% is generally too high due to missing processes). Countries with smaller FED, however, tend to deviate more from this target, which can be explained by the higher relative impact of both missing processes and errors in technological and economical assumptions[15] (e.g. SEC for processes, area for space heating). A detailed look at the iron and steel subsector shows that the overestimation via the bottom-up approach is structural (Fig. 15). At the process level, we found that this is mostly accountable to the energy demand of pig iron production (blast furnaces), which is thus overrepresented in this subsector. However, the resulting error regarding the temperature profile is likely to be minor, since blast furnaces use the vast majority of energy anyway. An overview of the two relevant dimensions country and subsector (Table 7, ordered by heating demand) confirms the impression that the bottom-up coverage[16] is better in countries and subsectors with higher overall heating demand (top left) than in smaller countries and subsectors (lower right). However, the 20 countries with the lowest energy demand only account for 15% of the total EU28 + 3 energy demand (Fig. 14).

Considering these results, we can assume that the process-based approach represents the real industrial structure of the countries and subsectors with the highest energy demand well enough to form a relevant picture of their temperature profiles. Differentiating processes thus enhances the country-specific patterns of industrial energy use and does not create unreasonable distortions of subsectoral energy demand compared to top-down sources. In smaller countries and subsectors, the added value of the process-based approach is reduced considerably. Furthermore, the approach reflects the chemical industry as a major but strongly heterogeneous energy user less accurately than other subsectors.

Results compared to selected end-use balances

Among the available end-use balances mentioned, we chose those of the UK and Austria as benchmarks for our results[17].

The UK's end-use balance ECUK (Department for Business, Energy and Industrial Strategy 2016a) comprises the dimensions subsector, end-use and energy carrier. Since the absolute values match neither those of Eurostat (2016b), nor our results nor those of the aggregated energy balance DUKES (Department for Business, Energy and Industrial Strategy 2016b) for reasons of methodology and scope, we only compare the relative shares of the end-uses. In the ECUK reports, these are high temperature (i.e. mainly iron and steel, non-metallic minerals), low temperature (i.e. mainly food, chemical, paper), and drying and separation (i.e. mainly paper). Detailed information on the definitions

[15] Smaller countries tend to have a less complete process portfolio. This has the effect of enhancing any errors in the assumptions about processes that do exist. Additionally, outliers of specific energy consumption are accorded higher weight.

[16] Note that Ireland (5) and Estonia (10) have very poor BU-coverage in Fig. 15, indicating inconsistency between the energy balance and production statistics.

[17] The German end-use balance 2012 is not a suitable candidate for comparison as it was compiled with the support of the same model we use here.

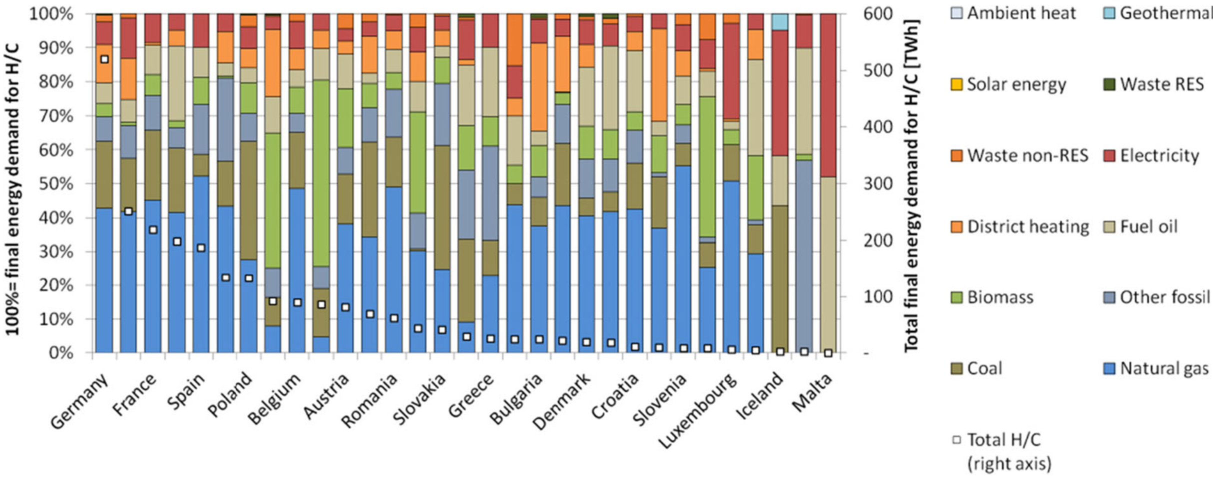


Fig. 9 Energy carrier share and final industrial energy demand for heating and cooling by country for the EU28 + 3 (model results based on Eurostat 2016b)

can be found in Department for Business, Energy, and Industrial Strategy (2016c). Tables 8 and 9 show the allocation we assumed for temperature levels and subsectors based on the information available there.

While the overall temperature share patterns of the studies are similar (high- and low-temperature-dominated subsectors, Fig. 16), we can observe differences in the medium-temperature range that are most likely caused by the approximate temperature allocation given in Table 8. The distinctions made between > 200 °C and 200–500 °C and 200–500 °C and > 500 °C are not very sharp at the edges, possibly creating overlaps due to different process definitions among studies. Additionally, there are notable anomalies in three subsectors: ECUK allocates no space heat to the food subsector; possibly due to installations generating both space heat and low-temperature process heat; and the ECUK-subsector *basic metals* (SIC 2007: 24) includes both iron and non-ferrous metals, which blurs their respective temperature profiles considerably. The third anomaly is the low-temperature share in the chemical subsector, which is much higher in ECUK. Based on the information available, we can only assume that this is mainly steam generation, which would cover a much wider temperature range than assumed in Table 8. Thus, the model results *might* be a plausible fit to the ECUK-temperature distribution. However, there is no certainty due to the methodological differences mentioned.

The Austrian end-use balance (Statistik Austria 2016a) is based on the national overall energy balance (Statistik Austria 2016b) and is, in most parts, directly comparable to Eurostat. The industrial subsectors are

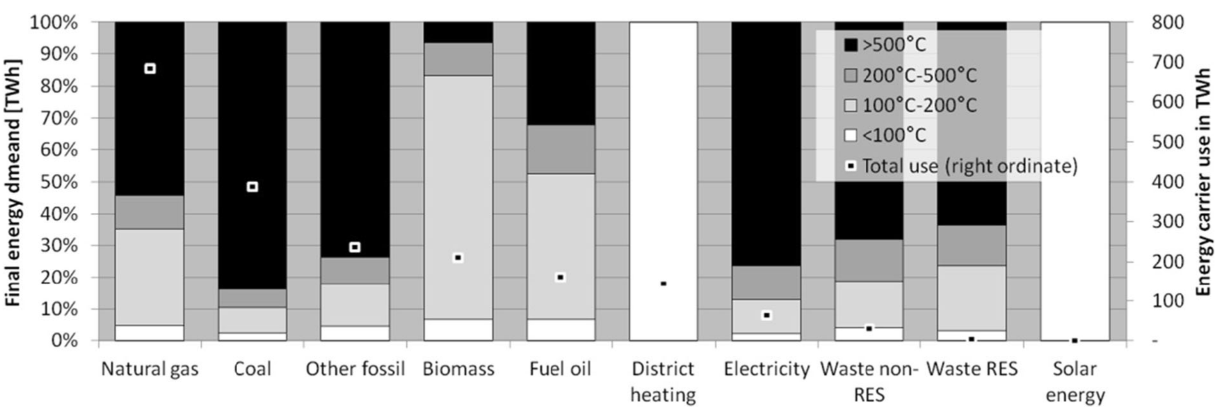
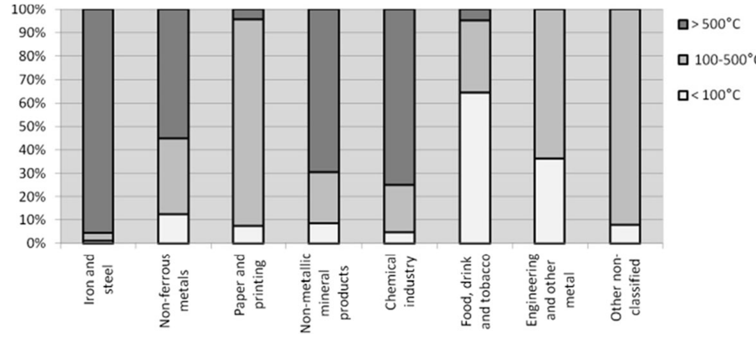


Fig. 10 Energy carrier use by temperature for industry in the EU28 + 3

Fig. 11 Subsidiary industrial subsector temperature distribution (used for energy demand not covered by bottom-up calculation, assumptions)



comparable, and the energy carrier categories used are similar to those of Eurostat (2016b). However, the iron and steel subsector has a different definition. Energy demand in blast furnaces is accounted as energy conversion in the overall energy balance (Statistik Austria 2016b) and not included in the detailed end-use analysis (Statistik Austria 2016a). For our purpose of comparing results, we assigned the entire demand of the energy carriers coke, coking gas and blast furnace gas balanced in the energy balance to *high-temperature demand* (> 500 °C), as these energy carriers are closely related to blast furnaces (and thus to high temperature). The

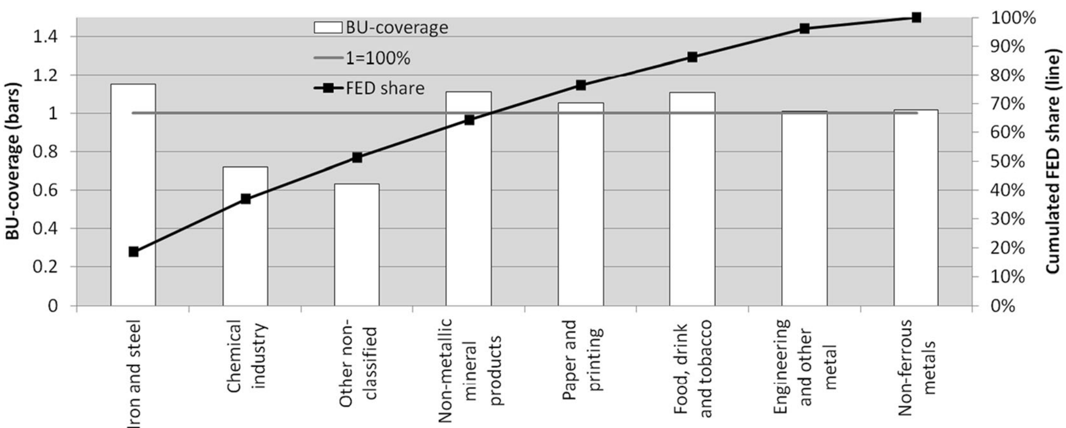


Fig. 12 Bottom-up coverage of industrial heating and cooling demand by subsector for EU28 + 3, 1 = 100%, right ordinate: cumulated FED share

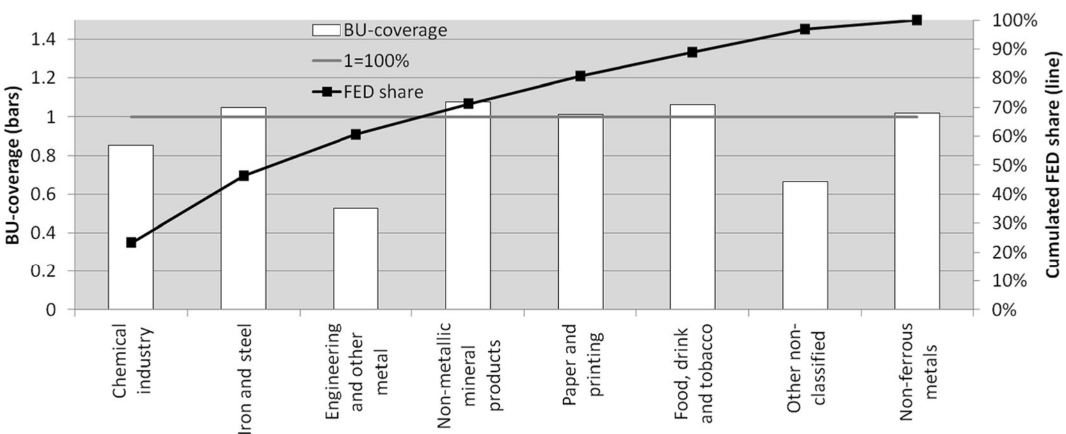


Fig. 13 Bottom-up coverage of industrial heating and cooling demand by subsector for Germany, 1 = 100%, right ordinate: cumulated FED share

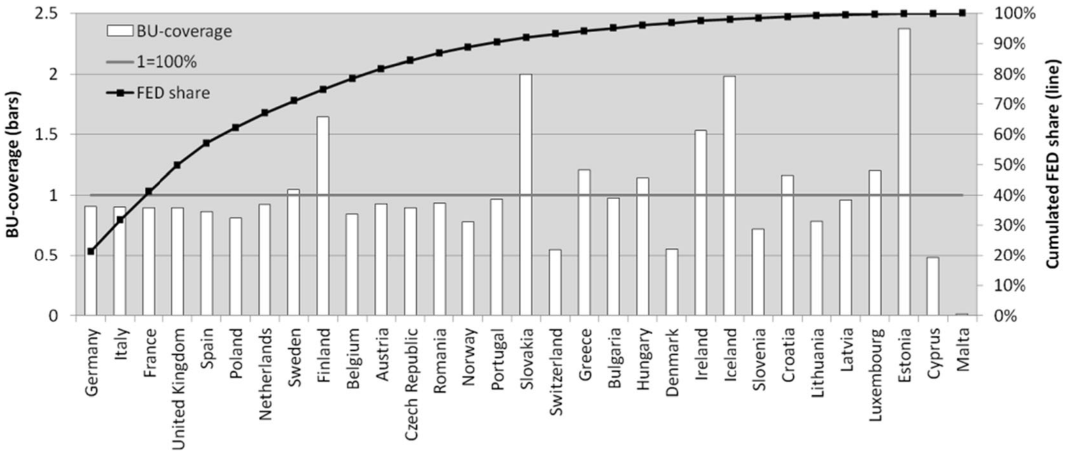


Fig. 14 Bottom-up coverage of industrial heating and cooling demand all subsectors by country, 1 = 100%, right ordinate: cumulated FED share

end-use balance adds the dimension of application (space heating, steam generation, furnaces and several non-heat uses) to the energy balance. We assigned these applications to temperature levels as shown in Table 10.

We observe general similarities for the subsectors of other industry, non-ferrous metals, iron and steel and paper, pulp and printing, although the temperature range suitable for steam generation is not explicitly defined (Fig. 17, compared to Table 10). This creates uncertainty about the profile for non-ferrous metals and non-metallic minerals. The Austrian end-use analysis does not include steam generation in these subsectors, thus excluding the temperature range of 200–500 °C *in our translation* (Table 10). Again, many differences in the compared temperature profiles might be explained by different allocations at the edges of temperature ranges. In 'Machinery and transport' as well as 'Other industry', we do not assume any high-temperature energy demand above 500 °C. However, these subsectors are not calculated in detail due to their strong heterogeneity even at process level. In the food subsector, space heating and steam generation seem to be connected (similar to what we observed in the UK).

Overall, we have to concede that the comparison with ECUK and with the end-use analysis created by Statistik Austria (2016b) does not necessarily support our results. We assume that the observed differences can be explained by different subsector and/or temperature/application definitions (as shown in Tables 8 and 10). This

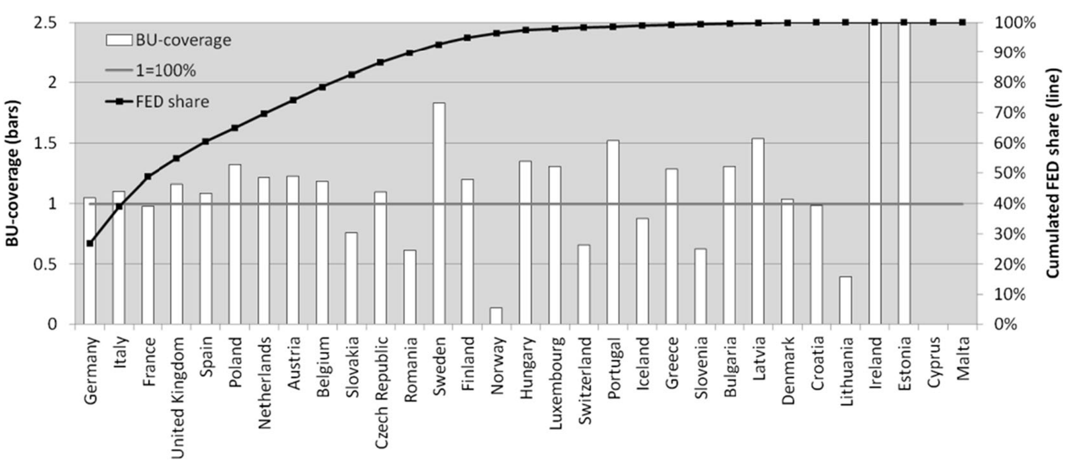


Fig. 15 Bottom-up coverage of industrial heating and cooling demand Iron and steel subsector by country, 1 = 100%, right ordinate: cumulated FED share; BU-coverage Ireland 5, Estonia 10

Table 7 Bottom-up coverage by country and industrial subsector (darker fields are closer to the expected value of approx. 0.9)

Country	Iron and steel	Chemical industry	Non-metallic mineral proc	Paper and printing	Non-ferrous metals	Food, drink and tobacco	Engineering and other me	Other non-classified	Country	Iron and steel	Chemical industry	Non-metallic mineral proc	Paper and printing	Non-ferrous metals	Food, drink and tobacco	Engineering and other me	Other non-classified
Germany	1.0	0.9	1.1	1.0	1.0	1.1	0.5	0.7	Switzerland	0.7	0.1	0.8	0.9	0.9	0.4	0.4	0.5
Italy	1.1	0.8	1.0	0.9	1.2	1.0	0.6	0.8	Greece	1.3	1.2	0.6	1.3	0.3	0.7	3.9	0.4
France	1.0	1.1	0.8	1.2	1.0	0.7	0.5	0.7	Bulgaria	1.3	0.5	1.3	0.5	0.8	1.5	1.2	0.8
United Kingdom	1.2	0.9	1.0	1.0	0.6	1.1	0.8	0.3	Hungary	1.3	1.8	1.0	1.1	0.1	1.0	0.7	0.9
Spain	1.1	0.3	0.9	1.2	0.7	1.3	1.2	0.5	Denmark	1.0	0.2	0.8	0.4	0.0	0.7	0.5	0.3
Poland	1.3	0.6	1.3	0.9	0.6	0.7	0.3	0.4	Ireland	4.9	0.4	1.1	1.6	0.0	1.1	0.5	0.3
Netherlands	1.2	0.2	1.3	1.2	1.2	0.7	1.3	0.7	Iceland	0.9	0.0	0.0	0.0	25.1	0.0	0.0	5.8
Sweden	1.8	0.2	1.8	1.0	1.1	1.3	1.1	0.2	Slovenia	0.6	0.4	1.4	1.0	0.4	1.1	0.5	0.4
Finland	1.2	0.8	2.0	0.9	0.8	2.6	5.1	0.8	Croatia	1.0	1.1	1.2	1.7	1.8	0.8	2.1	0.5
Belgium	1.2	0.4	1.0	1.0	1.4	0.7	1.0	0.5	Lithuania	0.4	0.8	0.7	1.2	0.3	1.1	1.2	0.6
Austria	1.2	0.8	0.9	1.0	1.1	1.1	1.1	0.4	Latvia	1.5	0.5	0.4	2.0	0.2	1.3	1.2	0.3
Czech Republic	1.1	0.6	1.2	0.9	1.1	1.2	0.8	0.5	Luxembourg	1.3	0.5	1.6	2.1	0.0	0.7	2.8	0.5
Romania	0.6	0.4	1.5	1.8	0.0	1.4	0.9	0.7	Estonia	10.0	0.5	0.6	1.0	0.2	1.2	0.7	0.3
Norway	0.1	0.8	1.2	1.1	1.0	1.1	1.1	0.3	Cyprus	0.0	0.3	1.0	0.6	0.6	0.9	0.9	0.2
Portugal	1.5	0.7	0.8	0.9	0.6	1.3	1.1	0.6	Malta	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Slovakia	0.8	0.9	1.8	1.2	1.6	3.9	3.6	3.5									

highlights the fact that a common methodology and definition of the included dimensions are required for comparability.

Results compared to recent publications

To further interpret the results of our approach, we compared them with recent publications (Naegler et al. 2015; Pardo et al. 2013). The work of Naegler et al. (2015) seems well suited to highlight the benefits of the detailed bottom-up data and the combination of empirical and modelling elements because they use a similar methodology and some of the same data sets (e.g. ISI 2013; Eurostat 2016b). Directly comparing the absolute energy demand of our approach and two data sets of Naegler et al. (2015) at country level, (Fig. 18) shows that, in general, data set 2 (DS-2) in Naegler et al. (2015) seems to be a closer match to our results. This is most likely due to the use of Eurostat energy balances in each case. The remaining differences are probably caused by different assumptions about non-heating/cooling use, which is especially relevant for electricity. Note that, although our results match the Eurostat energy balance (Eurostat 2016b) for 2012 at subsector and energy carrier level[18], there are considerable differences for some countries. In particular, there is a comparatively high absolute difference for Germany and France. Among the possible explanations, it seems plausible that this may be due to how space heating is accounted because of generally scarce data availability in industry regarding floor area as well as specific energy consumption. Another reason could be the use of electricity and how the share of electricity used for process heating is defined (see footnote on electrolysis in section *Temperature distribution*).

Table 8 Assumptions for temperature level relation ECUK/FORECAST

ECUK	FORECAST
High temperature	> 500 °C
Drying/separation	200–500 °C
Low temperature	< 200 °C
Space heating	Space heating
Refrigeration	All cooling
Other	–

[18] Naegler et al. (2015) claim to match the energy balance, too.

Table 9 Industrial subsector relation ECUK-FORECAST, (...) indicate shortened names

ECUK	FORECAST
Basic metals	Iron and steel, non-ferrous metals
Coke and refined petroleum products	Iron and steel, non-ferrous metals
Food products	Food, beverages and tobacco
Beverages	Food, beverages and tobacco
Tobacco products	Food, beverages and tobacco
Chemicals and chemical products	Chemicals
Non-metallic mineral products	Non-metallic minerals
Paper and paper products	Paper
Printing and publishing (...)	Paper
Machinery (...)	Machinery and transport
Motor vehicles (...)	Machinery and transport
Other transport equipment	Machinery and transport
Rubber and plastic products	Other industry
Textiles	Other industry
Wearing apparel	Other industry
Wood (...)	Other industry
Other manufacturing	Other industry
Electrical equipment	Other industry
Furniture	Other industry
Leather and related products	Other industry
Fabricated and metal products (...)	Other industry
Computer (...)	Other industry
Other mining and quarrying	Other industry

Table 10 Assumption on temperature level relation Statistik Austria (2016a) to FORECAST

Statistik Austria	FORECAST
Space heating	Space heating
Space generation	200–500 °C
Furnaces	> 500 °C

Figure 19 compares the temperature shares used by Naegler et al. (2015), Pardo et al. (2013) and our approach. While the temperature distribution shows a similar pattern for most subsectors, there is the general trend that our process-based approach shows lower demand below 100 °C than the subsector-based approaches. Most characteristic for the profiles is the relevance of high-temperature energy demand in minerals and metallic industries (e.g. blast furnace, cement kiln, melting furnaces), medium-temperature demand in paper and printing (steam for paper production and drying) and low-temperature demand in the food subsector as well as engineering and other metal and other non-classified industries. It should also be noted that we use a production-weighted average of country-specific temperature profiles in Fig. 19 for the FORECAST-data set (since the temperature profiles are actually used at process level). It can also be observed that both subsector-based approaches, Naegler et al. (2015) and Pardo et al. (2013), tend to agree more on the temperature shares; this is most apparent in the chemical and non-metallic industries. As mentioned before in the comparison with national end-use balances, the different temperature range definitions impede comparison.

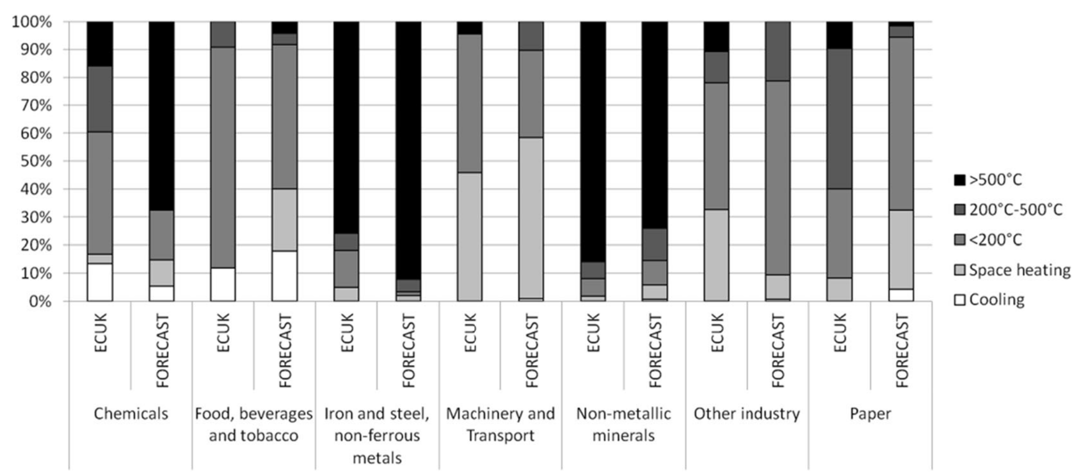


Fig. 16 Temperature shares by industrial subsector in the UK, ECUK and FORECAST (data from Department for Business, Energy and Industrial Strategy 2016a, FORECAST, own illustration, reallocated subsectors and temperature levels according to Tables 8 and 9)

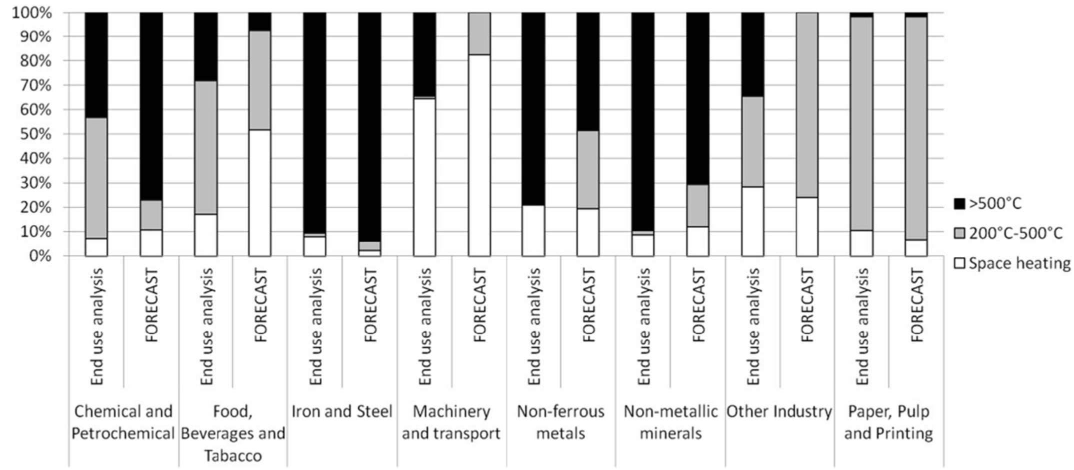


Fig. 17 Temperature shares by industrial subsector in Austria, end-use analysis (*Nutzenergieanalyse*, Statistik Austria (2016a)) and FORECAST

Methodological issues

The methodology relies on the availability of production data at process level which is sometimes a drawback, especially for smaller countries or those with an unusual industrial structure. In addition, analysing the bottom-up coverage shows that, process structure can be very diverse in some sectors (like chemicals), limiting the achievable share of bottom-up modelled industry (i.e. the *process equals process* assumption is violated or processes are neglected). For example, as of today, the ECHA (2016) lists over 15,000 products related to the chemical industry, of which 43 are produced in tonnages over 10 Mt per year. Even minor products or processes (in terms of energy demand) can still add up to a relevant share of the total energy demand. FORECAST's focus on energy-intensive industries, which necessarily neglects others, is thus both its strength (in terms of technological explicitness) and its weakness (in terms of completeness). Detailed knowledge about processes and their key indicators (specific energy demand, temperature profile, applicable energy carriers) is essential to

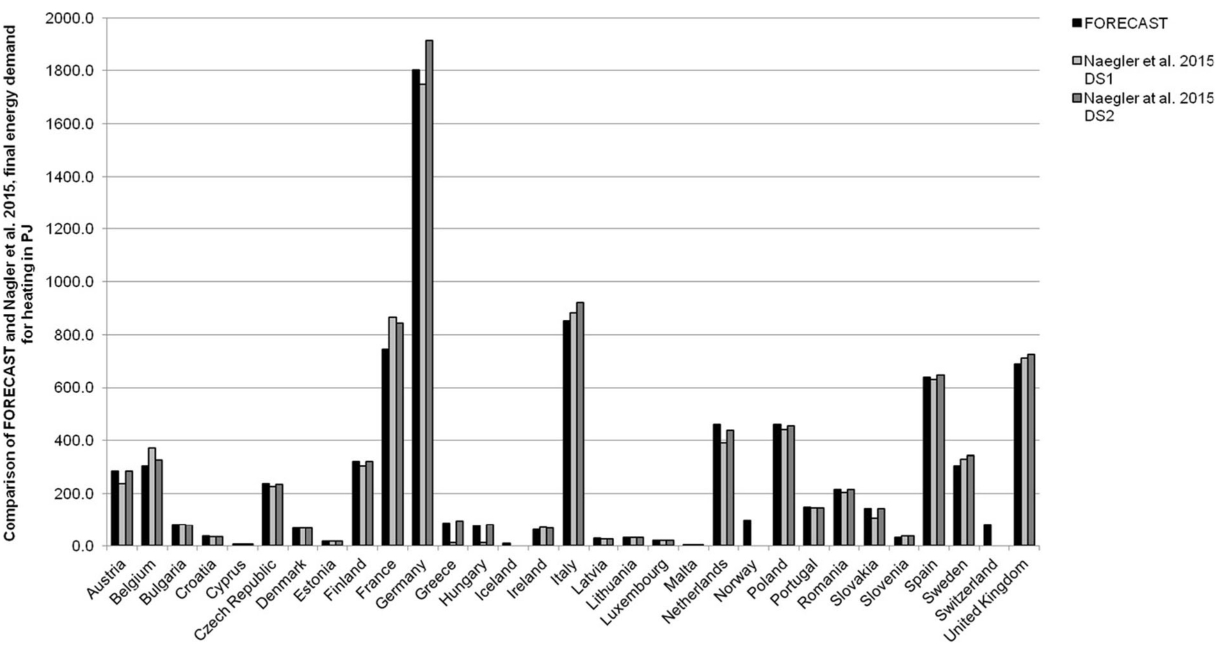


Fig. 18 Industrial final energy demand for heating, comparison of FORECAST (EU28 + 3) and Naegler et al. (2015) (EU28)

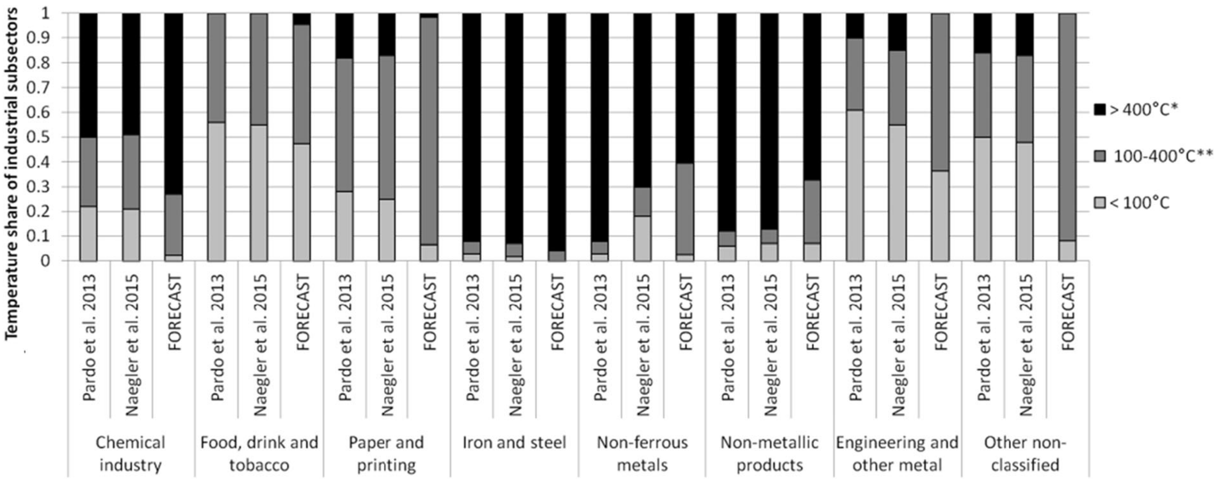


Fig. 19 Temperature share assumptions by industrial subsector: Comparison of Pardo et al. (2013), Naegler et al. (2015) and FORECAST; FORECAST: weighted average of processes in respective subsector, *FORECAST, > 500 °C; **FORECAST, 100–500 °C

achieve plausible results. Apart from very general challenges regarding energy efficiency-related indicators[19] (Patterson 1996), industrial energy demand is especially difficult to capture in terms of data availability. Specialised and small processes are particularly problematic that have relatively low importance in terms of absolute energy demand and have therefore not been studied in detail. Nevertheless, it remains questionable whether increased efforts in this regard can be justified in general. Fleiter et al. (2013) found that, for Germany, the processes included in FORECAST in its present state cover up to 86% of fuel demand, with the 20 largest processes already covering around 75% (see also Fig. 12). Further research should therefore focus on singular processes that are identified as particularly relevant in the countries (or subsectors) of interest.

Shortcomings in any of these fields (technological or economic data) result in a loss of precision, as energy demand that is not covered by the bottom-up approach is treated on an aggregated subsector level. This means that smaller processes that are not modelled in detail tend to be marginalised, even if they have greater relevance in certain countries with an unusual economic structure. This accords greater weight to the temperature distribution aggregated at subsector level, which can act as a fallback option to alleviate this drawback.

The basic assumption *process equals process*, while mostly sensible in general, ignores some differences in energy efficiency or other process characteristics that exist among countries with technological disparities (e.g. electricity use in the chemical industry is very different in France and Germany). Further research seems necessary on how technological differences among countries contribute to their process efficiency and specific energy demand, possibly linking them to more accessible data on a higher aggregation level.

Conclusions

Within this paper, we presented a methodology to disaggregate energy balances and add the dimensions *temperature level* and *end-use* using a bottom-up energy demand model. We investigated how including processes as the lowest level of differentiation in the production process adds to temperature profiles by comparing our results with previous work based on subsectors and introducing *bottom-up coverage* as a quality criterion for the methodology. In doing so, we generated an end-use balance for the EU28 + 3 in 2012 based on Eurostat's energy balance (Eurostat 2016b). Our approach of combining bottom-up data at process level with top-down energy balances and the results for the EU28 + 3 presented here show promise to support energy demand research and

[19] Patterson (1996) categorises 'persistent methodological problems': value judgements (e.g. what is useful energy), energy quality (e.g. enthalpy vs. exergy), boundaries (e.g. which input is considered to enter the energy balance and in what quality and state?), joint production (e.g. what is the main product of a process with multiple outputs and how to assign its energy use; combined heat and power production is the most popular example) and technical/gross efficiency.

policy design[20]. Compared to individual national energy and end-use balances, our approach has the main advantage that results are comparable due to a consistent methodology across all countries. It combines technologically explicit knowledge about processes and their individual properties needed to design targeted policies with real production values and available energy balances (Eurostat or national). As Naegler et al. (2015) point out, studying the processes applied in industrial subsectors demands details regarding their temperature profiles. We agree with their finding that a comprehensive estimation of the temperature levels of process heat demand is needed and made an effort to contribute to this (Table 1). The general approach that specific energy demand multiplied by production equals total energy demand is immediately evident, and the well-maintained database of processes and activities ensures high bottom-up coverage of many large and most smaller countries.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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[20] For example, by providing a more detailed picture of waste heat potentials, the use of temperature-dependent technologies like heat pumps and solar thermal systems, or by estimating the effect of differences in industrial structure on heating demand (via a country comparison).

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